

Rerank-by-Example: Efficient Browsing of Web Search Results

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Abstract. The conventional Web search has two problems. The first is that users' search intentions are diverse. The second is that search engines return a huge number of search results which are not ordered correctly. These problems decrease the accuracy of Web searches. To solve these problems, in our past work, we proposed a reranking system based on the user's search intentions whereby the user edits a part of the search results and the editing operations are propagated to all of the results to rerank them. In this paper, we propose methods of reranking Web search results that depend on the user's delete and emphasis operations. Then, we describe their evaluation. In addition, we propose a method to support deletion and emphasis by using Tag-Clouds.

Keywords: edit-and-propagate, tag-cloud, reranking, user-interface.

1 Introduction

A variety of ranking algorithms have been proposed and implemented. They are used in many Web search engines such as Yahoo! and Google. However, they sometimes do not return search results that adequately meet the user's needs as they are affected by two big problems in Web searches.

The first problem is that users' search intentions are diverse. For example, if a product named "A" is used as a query, one user might be seeking reviews of "A", whereas another user might be seeking information on how to buy "A". Moreover, the search intentions might be diverse even if the same user uses the same query. If a Web search engine returns the search results that do not satisfy the user's intention, the precision of the search results is considered low, because the user has to check more search results or modify the query. Instead, if the user could inform the user's search intention to the system while browsing the search results, the system might be able to return good search results for the user.

The second problem is that Web search engines return a huge number of Web search results, but the user usually checks merely the top 5 or 10 results. Therefore, almost all Web search results are not checked by the users. Moreover, there has been widespread usage of search engine optimization (SEO) techniques recently. Many companies and site owners spend much money on SEO in order to raise the search

rank order of their Web sites. As a result, currently, high ranked search results are often the result of applying SEO techniques and the accuracy and reliability of Web search results have decreased. This problem stems from the fact that it is burdensome for the user to check low ranked search results. Therefore, if we introduce reranking functions that the user can use easily, the user will be able to check a lot of good Web search results efficiently and the impact of SEO techniques will become weaker.

To solve these problems, in our past work, we proposed and implemented a reranking system based on edit-and-propagate operations [1]. In this work the system enables the user to edit any portion of a browsed page of Web search results at any time while the user is searching. Our system detects the user's search intention from the editing operations. For example, if the user deletes a part of the search results, our system guesses that "*this user does not want this kind of the result*". If the user emphasizes a part of the search results, our system guesses that "*this user wants more of this kind of result*". Our system propagates the users' search intention based on their editing operations to all search results in order to rerank them. In this way, the user can easily obtain optimized search results.

In this paper, we propose the three reranking methods of Web search results when the user performs the delete/emphasis operation to rerank them. Then, we provide the experimental results of these reranking methods. In addition, in order to perform delete/emphasis operations more effectively for the user, we propose and implement *MultiTagCloud*, which are generated by using Web search results.

2 Edit-and-Propagate Operations for Reranking

Our system reranks search results according to the type of operations users want to edit and the target that the user edits. The flow is as follows:

1. The user inputs a query to our system.
2. The system sends to the query to Web search engine.
3. The system receives the search results and shows them to the user.
4. The user checks the search results ranked by the search engine.
5. The user edits a portion of the search results by using editing operations.
6. The system detects the user's editing operations and guesses the user's search intention. Then the system reranks the whole search results according to this guess and shows the reranked search results to the user.
7. If the user is not satisfied with the reranked search results, go to 4.

The editing operations are as follows:

- **Deletion:** Deletion is an operation that indicates what types of search results the user does not want to obtain from the system.
- **Emphasis:** The role of an emphasis operation is opposite to the delete operation. Emphasis operation indicates what types of search results the user wants to obtain.

Search engines usually return title, snippet and URL as one of the search results to the user. If a user deletes/emphasizes a keyword or sentence from a title or snippet of

a search result, the system guesses that the user wants to degrade/upgrade some results which include the deleted/emphasized text. The system also guesses that the user wants to degrade/upgrade some results which include the topic about the deleted/emphasized text. For example, while browsing the Web search results of “Katsumi Tanaka”, if a user deleted “pianist” from a search result, the system guesses that the user has the intention to degrade the search results which are related to the topic about Katsumi Tanaka who is a pianist.

If a user deletes a URL or a part of a URL, the system guesses that the user wants to degrade the results which include deleted strings. If the user does not need the results whose domain is JP, he/she might delete “.jp” from a URL of the search results. If the user deletes a search result item, the system guesses that he/she wants to degrade the search results which are similar to the deleted item.

Figure 1 shows the implementation of our system when the user submits “kyoto” as the query. For this query, some of the high ranked search results are related to the Kyoto Protocol. If the user does not need information about the Kyoto Protocol, the user deletes “protocol” from the search result. After deleting a keyword, the system guesses the user’s intention and degrades the search results which include “protocol” in their titles or snippets. Then, the user receives reranked search results whose high ranked ones are related to Kyoto City or Kyoto University.

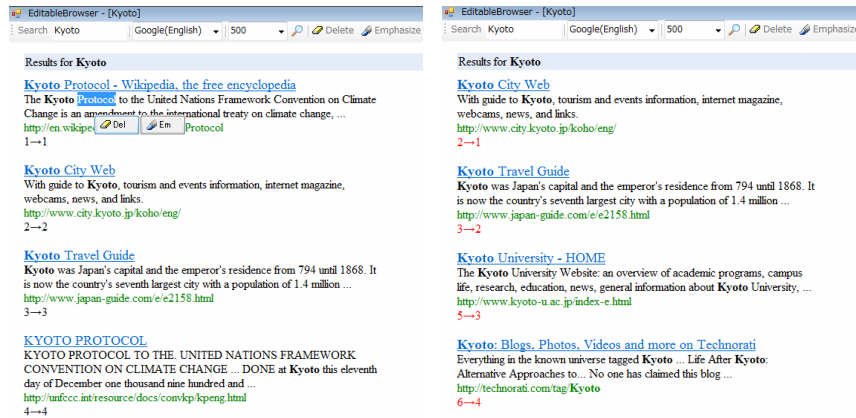


Fig. 1. Examples of the results produced by our system: The search results are reranked after deleting a keyword

3 Reranking Methods

Our goal is to estimate the user's intention when he/she uses editing operations and to reflect this intention in the search results page. For this purpose, we propose methods of reranking Web search results by deleting or emphasizing a keyword.

When the user inputs a query to our system, the system obtains the top N search results $r_1, r_2, \dots, r_N$ (r_i is the i_{th} search result item). Then, the system initializes the score of each search result item as:

$$Score(r_i) = N - i \quad (1)$$

(This formula allocates scores to search results depending on their search ranks.)

If the user deletes/emphasizes a keyword, the system re-calculates the scores of search result items and shows the search results in order of their scores to the user.

3.1 Selection Keyword Method

When the user deletes or emphasizes the term t , this method degrades or upgrades the search results including t . Our system calculates the score of r_i when it includes t in its title or snippet according to Equation 2.

$$Score(r_i)_{new} = Score(r_i)_{last} + type \times N \quad (2)$$

$$type = \begin{cases} 1 & \text{if edit operation is emphasis} \\ -1 & \text{if edit operation is deletion} \end{cases}$$

3.2 Extended Keyword Method

When the user inputs query like “*Katsumi Tanaka*” in order to seek information about Katsumi Tanaka who is a professor, he/she receives the search results which include not only a professor, but also a pianist, a poet, and so on. If a user deletes the term “*pianist*” in the search result, the system guesses his/her search intention as “the user wants to degrade the results about ‘*Katsumi Tanaka who is a pianist*’”. So when the user deletes “*pianist*”, it is better for the user to degrade the search results that not only include “*pianist*” but also include terms like “*piano*” or “*concert*” and so on. To accomplish this, we propose two methods called “*inner extended keyword*” and “*outer extended keyword*”.

If the user deleted/emphasized term t , both methods obtain terms $w_1, w_2, \dots, w_k$ as extended keywords. Then, for each term $t, w_1, w_2, \dots, w_k$, the system calculates the score of the search result items which include these terms in their titles or snippets according to Equation 2. $R(q)$ denotes the set of search result items of q . $W(q)$ denotes the set of terms which appear in $R(q)$. $TF(w, R(q))$ denotes the term frequency of $w \in W(q)$ in $R(q)$. In this work, by using Japanese morphological analyzer ChaSen [3], we extract proper nouns and general nouns as terms appear in the search results.

- **Inner Extended Keyword Method:** To obtain extended keywords, this method uses the term frequency of search results that include the term (we call this set of search result items as $R'(q)$). For each term $w \in W(q)$, the system calculates $TF(w, R'(q))$ and extracts the top k terms in order of this value as extended keywords.
- **Outer Extended Keyword Method:** This method uses the difference of term frequency between $R(q \text{ AND } t)$ and $R(q)$. For each term $w \in W(q)$, the system calculates $TF(w, R(q \text{ AND } t)) - TF(w, R(q))$. Then the system extracts the top k terms in order of this value as extended keywords.

For example, when the user deletes “*pianist*” in the search results of “*Katsumi Tanaka*”, the inner extended keyword method degrades the search results which

include “homepage”, “information”, “profile”, or “concert”. On the other hand, the outer extended keyword method degrades the search results which include “profile”, “concert”, “lesson”, or “piano” (if the system uses Google, $N=500$ and $k=4$).

4 Experiment

To evaluate our reranking methods, we investigated how the search results are reranked by deleting or emphasizing keywords using the reranking methods in Section 3. We prepared 15 search tasks and also prepared the correct answer sets for each search task.

In the keyword deletion task, the user inputs the prepared search query and submits the query to the search engine using our system. Then, the user checks the search results sequentially after receiving the search results. If the user reaches an unmatched search result, the user selects the keyword and deletes it. If the system detects the user's deletion, the system reranks the search results and shows the reranked search results to the user. Then the user also rechecks the search results from top to bottom. The user does this three times.

After each deletion, we calculated the precision of the ranking and the number of correct search results which the system incorrectly degraded. We compared the *selection keyword method (SKM)*, *inner extended keyword method (IEKM)*, and *outer extended keyword method (OEKM)* in this task.

In the keyword emphasis task, the user inputs the prepared search query and submits the query to the search engine using our system. Then, he/she checks the search results sequentially after receiving the search results. If the user finds a keyword which the user thinks is relevant, the user selects the keyword and emphasizes it. If the system detects the user's emphasis operation, the system reranks the search results and shows the reranked search results to the user. Then the user rechecks the search results from top to bottom. This action is performed once, because the keyword emphasis task is so effective that after second emphasis almost all the top 20 reranked search results become correct.

After each emphasis, we calculated the precision of the ranking. We compared the *SKM*, the *IEKM*, and the *OEKM* in this task.

If the reranked all top 20 search results are correct, the user stops the operation.

Figure 2 shows the average precisions of the top K ($K=1, 2, \dots, 20$) in the search results of each deletion. Figure 3 shows average precisions of each emphasis. Figure 4 shows average initial order of lowest original matched search results in top K by using *SKM*. This figure shows how low rank of the matched search results are upgraded as the user performs the emphasis operation. Table 1 shows the numbers of matched search results which are removed from the top 20 by the deletion. Number of total deletions means the total number of search results which are removed from the top 20 by three delete operations. Number of error deletions means the number of the matched search results which are removed from the top 20 by three deletions. Error ratio means the ratio of the total deletion number and the error number.

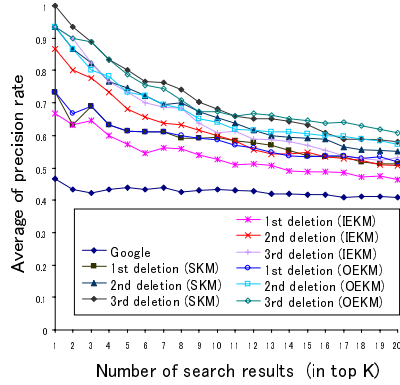


Fig. 2. Average precision obtained after each deletion

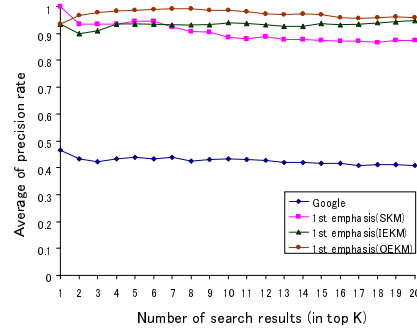


Fig. 3. Average precision obtained after each emphasis

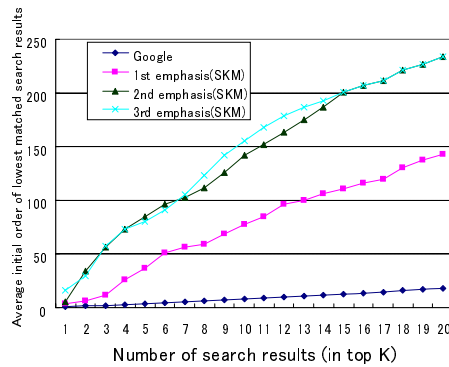


Fig. 4. Average initial order of lowest matched search results in top K

Table 1. Ratio of matched search results removed from the top 20

	# of total deletions	# of error deletions	Error ratio
SKM	127	2	1.57%
IEKM	191	20	10.47%
OEKM	208	35	16.83%

4.1 Discussion

Figure 2 shows that the number of relevant search results in the top 20 increases after each delete operation in all reranking methods. We note that the effect of the first deletion is biggest and the effect of the deletion becomes smaller as the user repeats deletions.

The effect of the *IEKM* is smallest and the difference between the effectiveness of the *OEKM* and that of the *SKM* was relatively small. In particular, if we assume that users usually check only the top 10 search results, both reranking methods have the same effectiveness. As shown in Table 1, the main reason for this is that although the two extended keyword methods degrade more search results than the *SKM* does, they also degrade more relevant search results. As shown in Section 3.2, when the user deletes “*pianist*” from the search results of “*Katsumi Tanaka*”, the *OEKM* degrades

the search results which contain terms related to “*pianist*”. In contrast, the *IEKM* degrades the search results which contain “*information*”. The search results which are related to the Katsumi Tanaka who is a professor have frequent co-occurrences with “*information*”. Therefore, when the system degrades the search results which contain “*information*”, many relevant search results are degraded. The *OEKM* also obtains popular terms or terms which appear in multiple topics although the *OEKM* obtains fewer such terms than the *IEKM*. We need to solve this problem to increase the effect of the two extended keyword methods.

The effect of emphasis operations is much bigger than that of delete operations. We note that the first emphasis operation is effective, by which all top 20 search results are relevant in more than two-thirds of the queries.

In the emphasis operation task, the *OEKM* outperformed the *SKM* and the *IEKM*. In the delete operation task, the fact that the extended keyword methods often obtain popular terms like “*information*”, which are related to multiple topics, lowers their accuracy. In the emphasis task, however, if the term which the user emphasized directly is related to relevant search results, this problem does not occur. We think the extended keyword methods outperformed the *SKM* because of this reason.

As shown in Figure 4, our system can pick up very low ranked relevant search results as the user performs the emphasis.

5 Supporting Deletion and Emphasis by Using MultiTagCloud

A tag-cloud is used as a visual depiction of tags. In a tag-cloud, more frequently used tags or more popular tags are shown bigger or more emphasized. The widely known use of tag-clouds is made by flickr [4]. Flickr displays many kinds of tags that users tag to photos by using a tag-cloud. One of the biggest merits of tag-clouds is that users can instantly check the popular tags and easily look over many tags.

We therefore applied this interface model to the Web search results page in order to support the use of the delete and emphasis operations for users.

5.1 Method of Generating MultiTagCloud

The general tag-cloud is a list of tags that users tagged. Our tag-cloud is a list of keywords that appear in Web search results. Moreover, in order to visualize how the contents of the Web search results are distributed on the page, we split search results into several groups, and for each of the groups we generate a tag-cloud. The system shows these tag-clouds simultaneously as a *MultiTagCloud*. The process of creating a *MultiTagCloud* is as follows:

1. The system splits a set of the search result items $R=\{r_1, r_2, \dots, r_N\}$ in page p , ordered by their scores, into n groups. (We call this split group $R'=\{R_1, R_2, \dots, R_n\}$)
2. For each $R_i \in R'$, we generate a tag-cloud according to the following steps.
 - (a) By doing a morphological analysis on the titles and the snippets of the search result items in R_i , the system obtains the set of keywords T_i , and calculates the term frequency of keyword $T \in T_i$

- (b) The system finds the top k frequent keywords, and then adds them to a tag-cloud in appearance order in R_i .
 - (c) When the term frequency of a keyword in a tag-cloud exceeds the threshold $\theta_i (i=1, 2, 3, \dots)$, the system modifies the font size of the keyword in it.
3. The system simultaneously shows all the tag-clouds as a *MultiTagCloud*.

5.2 Example of System Output

Figure 5 shows an example of the output produced by our system. The normal search results are displayed on the left side of the system. The *MultiTagCloud* is displayed on the right side of the system. In Figure 5, the system displays the top 500 search results, with *MultiTagCloud* consisting of tag-clouds of the 1st to 100th, 101st to 200th, ..., 401st to 500th search results. For each tag-cloud, the system displays keywords whose term frequency is within the top 30. The users can use editing operations on any of the keywords in the tag-clouds as well as those in search results. Therefore, the user can rerank search results by deleting or emphasizing any keywords in the *MultiTagCloud*. According to the user's operations, the system reranks the search results using the methods in Section 3, and then the system shows the reranked search results and the re-generated *MultiTagCloud* (right of Figure 5).

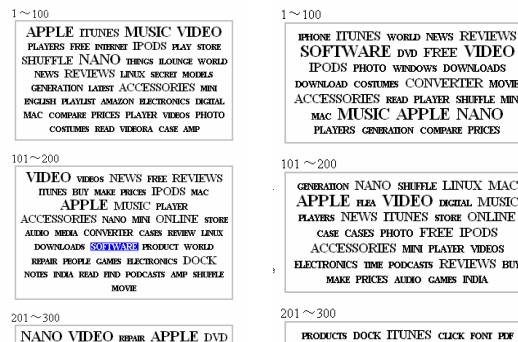


Fig. 5. Example of *MultiTagCloud*. The *MultiTagCloud* is re-generated depending on the reranked search results (Right).

5.3 Effectiveness of MultiTagCloud

By showing *MultiTagCloud* with the search results, the user can easily and quickly understand what keywords are influential.

We can visually understand that if we delete a keyword displayed in a large font, many search results will be reranked to the bottom and that if we emphasize a keyword displayed in a small font, a few sparse search results will be gathered and reranked to the top. For these reasons, showing tag-clouds enables users to know how to modify the search results when they delete or emphasize a keyword. Users can more easily find keywords they need or do not need in a tag-cloud than in the conventional search results. Moreover, by using a *MultiTagCloud*, the user can also

see the propagation of keywords. In Figure 5, when the user submits “*iPod*” and then emphasizes “*software*” in a *MultiTagCloud*, keywords related to the “*software*” such as “*iPhone*”, “*iTunes*”, “*Free*”, and “*Download*” come to appear at the top of the *Multi-Tag-Cloud*.

6 Related Work

Broder [2] and Rose and Levinson [8] have classified the goals of a user searching the Web into three categories: *navigational*, *informational*, and *transactional/resource*. There are also many studies on automatic classification of user goals by using queries [5, 6]. These works focused on the recognition of the users' search intentions by analyzing their queries. However, it is difficult to recognize such intentions just by analyzing the inputted queries because even if the same user uses the same query in a Web search, the user's search intention might depend on the time and situation. Our goal is to estimate the user's search intention through interactions between the user and the system, and to reflect this intention in the Web search results page.

Yahoo! Mindset [11] allows the user to specify his/her search intentions to the system. It weights each Web page as ‘more research’ or ‘more shopping’. The user can specify that his/her search intention is research or shopping by using a slide bar; then the system re-ranks the search results according to the weight. However, in this system, the user cannot re-rank the search intentions without prepared factors.

There are many reranking methods. Relevance feedback [9] is the most popular retrieval system to retrieve documents using the user's feedback. Relevance feedback is based on a vector space model [10]. Non-relevance feedback [7] is a retrieval method for documents. It uses only non-relevant documents to find the target documents from a large data set of documents. We think that we can apply both of the relevance feedback and non-relevance feedback to optimizing searched results.

7 Conclusion and Future Work

We proposed reranking methods for users to delete or emphasize keywords and did detailed evaluation tests. We showed the effectiveness of edit-and-propagate operations for reranking Web search results. In the future, we plan to compare the effect of our reranking methods with that of a query expansion which uses Boolean operators such as AND and NOT. We also plan to devise more advanced reranking methods because reranking methods proposed in this paper are simple algorithms.

Moreover, we proposed the use of the *MultiTagCloud* in order to support the delete and emphasis operations.

In this work, we only implemented reranking by keyword deletion and emphasis. In the future, however, we plan to implement reranking by deletion and emphasis of search results item or sentence in a snippet. In addition, we would like to introduce other operations such as drag-and-drop, and replace.

In this paper, we only focused on reranking. However, we think that the user may have many other intentions in browsing search results pages. For example, the user may want to modify a snippet which is generated by the search engine in order to

check another part of the original content which does not appear in the snippet. Users might also want to compare multiple targets using Web searches. We will attempt to support such a comparison by introducing editing operations.

Moreover, we can apply edit-and-propagate operations to browse other types of Web pages. For example, while browsing a bulletin board, users might want to delete improper messages or spam in order to increase readability, or emphasize a term in order to check topics quickly. We will develop such an editable browser for many kinds of Web pages.

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